

HomeService OS

4-Member Project Work Division & Development Plan

Technology: HTML, CSS, JavaScript, Python Flask, MySQL, MySQL Workbench

AI Agent: Planned for a later phase after the core product is stable.

Updated from the previous 3-member plan to 4 members. Members 1–3 own functional modules; Member 4 focuses exclusively on frontend UI/UX design across the complete product.

1. Final Team Division

Member	Main Responsibility
Member 1	Customer + Home + Appliances + Move Mode
Member 2	Services + Technicians + Nearby Matching + Booking
Member 3	Admin + Statistics + Payments + Reviews + Notifications
Member 4	Frontend UI/UX Design — All Modules

2. Member 1 — Customer + Home + Appliances + Move Mode

Responsibilities

- Customer registration, login, logout and profile.
- Home management: add, edit, delete homes; multiple homes; current/previous home.
- Appliance management: add/edit/delete appliances, details and warranty information.
- Customer service history and maintenance records.
- Move Mode: new address, appliance selection, required-service identification and technician matching request.

Backend

- Develop Flask routes/APIs for customer, home, appliance, service-history and move-request functionality.

Database

- customers
- homes
- appliances
- move_requests
- move_appliances

3. Member 2 — Services + Technicians + Matching + Booking

Services

- Service categories, details, estimated price, required skill and duration.

Technicians

- Registration, profile, skills, service areas, availability and experience.

Nearby Matching

- Use customer location, technician location, distance, skill match, availability and rating.

Booking & Execution

- Create booking, accept/reject, scheduling, cancellation and completion.
- Status flow: Pending → Assigned → Accepted → Scheduled → In Progress → Completed.

Database

- technicians

- services
- technician_skills
- technician_availability
- service_requests
- bookings
- service_history

4. Member 3 — Admin + Statistics + Payments + Reviews + Notifications

Admin

- Customer, technician, service, booking and complaint management.

Payments

- Payment records, service cost, additional cost, total, payment status and payment method.

Reviews

- Customer ratings, technician ratings, reviews and rating statistics.

Notifications

- Booking confirmation, technician acceptance, service completion, maintenance reminders and review requests.

Statistics & Analytics

- Customer, technician and admin analytics.
- Mean, median, mode, variance, standard deviation, quartiles, correlation, regression and appropriate hypothesis testing.

Database

- users
- payments
- reviews
- complaints
- notifications
- maintenance_reminders

5. Member 4 — Frontend UI/UX Design ONLY

Main Responsibility

- Focus exclusively on frontend visual design and user experience for the complete application.
- Use HTML, CSS and JavaScript.
- Do not own Python Flask, MySQL or backend logic.

Common Design System

- Color palette, typography, buttons, cards, forms, tables, modals, alerts, navbar, sidebar and footer.
- Maintain consistent spacing, layouts, icons and responsive behavior.

Customer UI

- Customer dashboard, home management, appliances, service history, bookings and Move Mode.

Technician UI

- Technician dashboard, profile, service requests, nearby requests, availability, booking management and service status.

Admin UI

- Admin dashboard, users, technicians, services, bookings, payments, reviews, complaints, notifications and analytics.

Statistics UI

- Statistics cards, charts, tables, filters and report presentation.

Responsive Design

- Desktop, laptop, tablet and mobile layouts.

6. How Member 4 Works With Members 1–3

Member 4 does not create a disconnected frontend. Member 4 designs/builds the UI for each module; the respective module owner connects Flask routes, database and business logic.

Frontend UI	Functional Owner
Customer/Home/Appliances/Move Mode	Member 4 UI + Member 1 functionality
Services/Technicians/Matching/Booking	Member 4 UI + Member 2 functionality
Admin/Payments/Reviews/Statistics	Member 4 UI + Member 3 functionality

7. GitHub Project Structure

```
HomeServiceOS/
■■■■ app.py
■■■■ config.py
■■■■ requirements.txt
■■■■ README.md
■■■■ database/schema.sql
■■■■ routes/
■ ■■■■ auth.py
■ ■■■■ customer.py
■ ■■■■ technician.py
■ ■■■■ booking.py
■ ■■■■ admin.py
■■■■ templates/
■ ■■■■ login.html
■ ■■■■ customer/
■ ■■■■ technician/
■ ■■■■ admin/
■■■■ static/
■ ■■■■ css/
■ ■■■■ js/
■ ■■■■ images/
■■■■ utils/
■■■■ database.py
■■■■ auth.py
```

8. Folder Responsibility

Area	M1	M2	M3	M4
Customer pages	Functionality			UI/UX
Technician pages		Functionality		UI/UX
Admin pages			Functionality	UI/UX
CSS / visual design				Primary owner
Flask backend	Customer	Technician/Booking	Admin/Analytics	—
Database	Customer tables	Service tables	Admin tables	—

9. Development Phases

Phase 1 — Planning — All 4: requirements, architecture, database planning, UI planning and GitHub setup.

Phase 2 — UI Foundation — Member 4: common design system and page layouts. Members 1–3: backend architecture and database preparation.

Phase 3 — Core Development — Member 1: Customer/Home. Member 2: Services/Technicians/Booking. Member 3: Admin/Statistics/Payments. Member 4: frontend UI for all modules.

Phase 4 — Integration — All 4: connect frontend, Flask APIs and MySQL; resolve integration issues.

Phase 5 — Testing — All 4: test login, customer flow, technician flow, booking, matching, Move Mode, admin, statistics, payments, reviews and notifications.

Phase 6 — Deployment — All 4: prepare and deploy the completed product.

Later — AI Agent — After the core product is stable, all 4 can contribute. Member 4 can focus on AI interface/chat UI while Members 1–3 provide module tools and backend integration.

10. Final Division

- Member 1 → Customer + Home + Appliances + Move Mode + related backend/database.
- Member 2 → Services + Technicians + Nearby Matching + Booking + related backend/database.
- Member 3 → Admin + Statistics + Payments + Reviews + Notifications + related backend/database.
- Member 4 → Frontend UI/UX Design for ALL modules using HTML, CSS and JavaScript.
- All 4 → Integration + Testing + Deployment.
- Later → AI Agent.